



Chihyeon Yun

AI Researcher & Engineer

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- ◆ **Multimodal Auto-Editing (M.S. & First Author):** Designed visual speaking-state and audio diarization late-fusion frameworks, utilizing Beam Search decoding to estimate optimal camera cuts (achieved 90.10% agreement with experts).
- ◆ **Spatial AI & Metaverse (ETRI R&D):** Engineered high-accuracy indoor Visual Positioning Systems (VPS) for cyber-physical space registration, and designed Visual-Language Model (VLM) zero-shot anomaly detection pipelines for real-time video semantics.
- ◆ **3D Reconstruction & VLA Robotics (PoC):** Coupled 2D MT-DUNet and Beer-Lambert radiative 3DGS for coronary artery 3D reconstruction under extreme memory optimization, and integrated SmoIVLA control in Unity ROS simulation.

Research & Career Timeline

2025.02 - 2025.06

Electronics and Telecommunications Research Institute (ETRI)

Post-Master's Research Fellow at Hyper-Realistic Metaverse Lab. Worked on indoor visual positioning (VPS) coordinates mapping and VLM semantic anomaly detection. Managed joint government R&D milestones.

2023.03 - 2025.02

Kyungpook National University

Master of Science in Artificial Intelligence (GPA: 4.11/4.3). Conducted research in Multimodal Language & Cognition Lab on audio-visual late-fusion, speech diarization, and generative audio diffusion.

Core Tech Stack

Deep Learning & Audio

- PyTorch
- Transformers
- DiffWave
- WhisperX
- Pyannote.audio
- Librosa

Vision & Robotics

- 3DGS
- Visual SLAM/VPS
- MT-DUNet
- SmoIVLA
- ROS/ROS2
- Unity

Medical Imaging

3D Reconstruction

LCA CAG 3DGS Reconstruction & Dashboard

Volumetric 3D reconstruction system recovering coronary artery structure from sparse 2D X-ray angiograms.

- ◆ **MT-DUNet (DCNv2)**: Adapts to complex vessel tortuosity, simultaneously predicting masks and 25 branch labels.
- ◆ **Class-Constrained Triangulation**: Eliminates ghost points via label matching (4.5x denser centerline seed).
- ◆ **Beer-Lambert 3DGS**: Solves X-ray projection opacity and radius profiling using differentiable rendering.

PyTorch, 3DGS, Deformable Conv, HTML/JS

AI Agent

RAG Systems

Chihyeon.ai Personal RAG Assistant

Interactive AI secretary bot resolving developer credentials, CV, and academic queries in real time.

- ◆ **High-Fidelity Parsing**: Leverages LlamaParse & Jina Reader API to extract formulas, tables, and web content.
- ◆ **Secure Serverless Proxy**: Uses Vercel Functions to encrypt API credentials and bypass region locks.
- ◆ **Vector Storage**: Fast semantic matching over high-dim Pinecone space powered by Gemini 2.5 Flash.

Gemini 2.5 Flash, Pinecone, Vercel, Python

Robotics

Simulation

Unity Pick & Place SmolVLA Integration

End-to-end integration of the SmolVLA (Vision-Language-Action) model within a virtual physics environment.

- ◆ **Vibe Coding Paradigm**: Worked with LLMs to prototype network interfacing, accelerating development.
- ◆ **Communication Pipeline**: Built ROS-Unity bridges linking virtual cameras and joint torques.
- ◆ **Sim-to-Real Control**: Feds camera feeds to VLA models for high-level spatial coordination logic.

SmolVLA, Unity, ROS, Docker, PyTorch

JOURNAL SUBMISSION (PENDING / FIRST AUTHOR)

"Seamless Multi-Hypothesis Video Editing Using Deep Multimodal Analysis"

Fuses active speaker detection (Swin Transformer) and diarization (WhisperX/Pyannote) visual-audio streams to align channels, decoding edits via a customized Beam Search logic matching 90.10% of expert visual pacing.

Key Stack: Speaker State, Multi-Hypothesis, Beam Search, Swin-T, Python

SCIENTIFIC DATA (NATURE PORTFOLIO, SCIE, 2024 / CO-AUTHOR)

"A Military Audio Dataset for Situational Awareness and Surveillance"

Curated and open-sourced 8,075 military audio clips (MAD) representing 7 hazard classes for security surveillance, designing raw audio preprocessing filters and evaluating AST models achieving 91.07% classification accuracy.

Key Stack: Audio Spectrogram Transformer (AST), Mel Filterbank, Librosa

NEURIPS 2023 DGM4H WORKSHOP (INTERNATIONAL CONFERENCE / CO-AUTHOR)

"Adversarial Fine-tuning using Generated Respiratory Sound"

Tackled medical data scarcity via DiffWave diffusion synthesis, training deep classification structures with domain-adversarial fine-tuning (AFT) loops to bridge synthetic and real sample sets, increasing minority class recall by 26.58%.

Key Stack: DiffWave Audio Diffusion, Adversarial Fine-Tuning (AFT), PyTorch

KCC PROCEEDINGS (DOMESTIC CONFERENCE / BRONZE PRIZE)

"CutMix-SupCon: Contrastive Learning for Multimodal Emotion Recognition"

Designed latent-level CutMix feature interpolation to merge KLUE-RoBERTa (text) and wav2vec 2.0 (audio) embeddings, optimizing Supervised Contrastive Learning (SupCon) for cosine similarity alignment on minor classes.

Key Stack: wav2vec 2.0, KLUE-RoBERTa, Supervised Contrastive (SupCon)